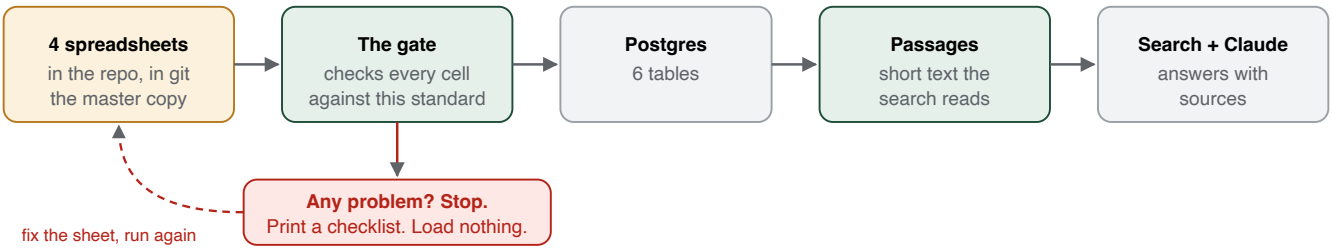
Houston Zoo data standard

How the four spreadsheets will be stored, checked, cleaned, and loaded

Rice Apps, Houston Zoo team · Working decisions from October 24, 2026 · Covers the four exports dated through October 15, 2026

The plan in one picture

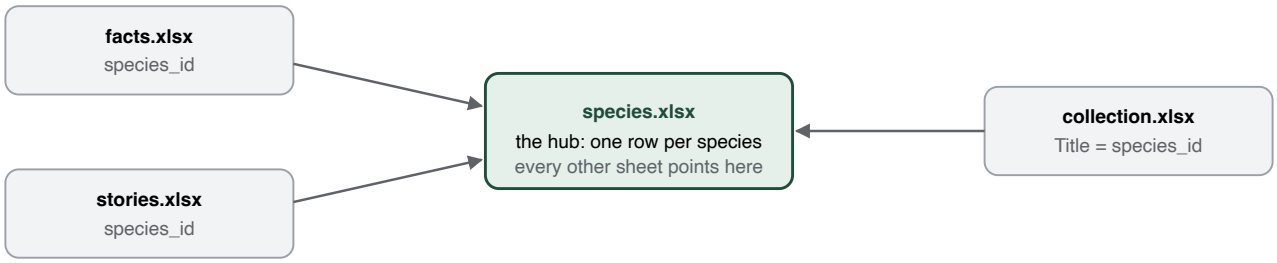


Five decisions that drive everything

- The four spreadsheets in the project folder are the master copy. We edit them, we commit them to git, and SharePoint is no longer a source.
- The data is cleaned once, by hand, against the rules in this document. The checklist in Appendix A says exactly what to fix.
- The ingest is a gate. It checks and refuses; it never guesses or repairs. The only thing it fixes on its own is invisible characters (stray spaces, tabs, carriage-return markers) that nobody can see in Excel.
- No new id columns. Rows are identified by what is already there: the species id, the fact id, the species plus story text, and the species plus house name. One column is added, `public`, because nothing in the data carries that meaning today.
- The species sheet is the only place a species name is spelled. Where the Collection sheet repeats a name, it has to match exactly.

The four sheets

File in `backend/data/`	One row is	Rows now	Identified by
species.xlsx (was "Natural History")	a species or taxon group	1,262	Title, the species id
facts.xlsx	one fact about a species	2,227	FactsID
stories.xlsx	one keeper story about a species	354	species id + the story text
collection.xlsx (was "Individual Animals")	one living animal	202	species id + house name



Stories and facts never point at an individual animal. A story names an animal only as a word in its text.

Rules that apply to every sheet

- The header row must contain exactly the columns listed below, in any order. A missing or renamed column stops the run.
- Spaces at the edges of a cell, tabs, non-breaking spaces, and `_x000D_` markers are removed silently. Nothing else is.
- Columns with a fixed list of values accept any capitalization (`mammal` and `Mammal` both pass) but nothing outside the list.
- True/false columns hold `T` or `F`. Blank is not allowed in those columns.
- Every species id in the facts, stories, and collection sheets must exist in the species sheet.
- A blank cell in any other column means "not recorded". The passage simply leaves that line out.

Sheet by sheet

species.xlsx

Column	Rule
Title	Required. A whole number. Unique.
scientific_name, common_name	Both required. The pair must be unique. Generic names like "Coral" are fine as long as the scientific name differs; passages print both, as "Coral (Porites)".
animal_class	One of: Mammal, Bird, Reptile, Amphibian, Fish, Invertebrate.
department	One or more from the department list (Appendix C), separated by semicolons.
location, facility_name	From the places list (Appendix C).
iucn_status	One of the nine IUCN categories (Appendix C). Trend notes such as "population decreasing" move into conservation_general.
diet	One of: Carnivore, Herbivore, Omnivore, Insectivore, Frugivore, Piscivore, Granivore, Filter feeder. Menus like "hay, grain, produce" belong in diet_zoo.
development_type	One of: Gestation, Incubation, Live birth, Ovoviviparous, Larval. Details like "4-5 eggs" belong in development_period.
venomous, sexually_dimorphic, ssp, handling, in_hzi_collection	T or F.

Everything else	Free text, kept as written: range, threats, conservation_general, conservation_hzi, conservation_references, development_period, birth and adult sizes, parental_care, sexual_maturity, median_life_expectancy, diet_zoo, diet_wild, habitats, description, references, range_map.

habitats and diet_wild looked like comma-separated lists at first, but half the cells are sentences. They stay free text.

facts.xlsx

Column	Rule
FactsID	Required. A whole number. Unique. (187 rows currently share the id 1; they get fresh numbers continuing from the highest id.)
species_id	Required. Must exist in species.xlsx.
summary	Required. Not the word NULL. No two facts with the same text under the same species.
priority	Optional whole number. Kept, not used, until the zoo tells us what it means.

stories.xlsx

Column	Rule
species_id	Required. Must exist in species.xlsx.
body	Required. The same text may appear under several species (one husbandry blurb is attached to 30 reptile species on purpose). The same text twice under one species is a duplicate and is refused.
StoriesID	Ignored. It is SharePoint's row number, 43 are blank and two repeat, so it cannot serve as a key.

collection.xlsx

Column	Rule
Title	Required. The species id. Must exist in species.xlsx.
Scientific Name, Common Name	Must match the species sheet exactly for that id. This is what catches an animal filed under the wrong species.
House Name	Required. Unique within the species. Written as a name, not in capitals. A nickname is allowed in one shape only: `Thailand (Thai)`.
Sex M,F,Unk	One of: male, female, unknown.
Date of Birth/Hatch	`mm/dd/yyyy`. Unknown parts are `xx`. A leading `~` means approximate. Blank means unknown. See the next section.
Current Habitat	From the places list (Appendix C).
Previous Institutions, Additional Previous Institutions	Entries like `Denver Zoo (birth); Fort Worth Zoo (lived)`. Only the tags (birth) and (lived). An animal born here reads `Houston Zoo (birth)`, which is what the embargo rule looks for.

Notes	Free text: weights, bands, markings, injuries.
public	New column. T or F. Default T. F hides the animal, its stories, and its place in the roster from any public-facing use.

Who fixes what

Two lists. The left one is the editing job in Excel, done once now and then kept up as the sheets change. The right one is what the parser does on every run, so nobody has to.

We edit the Excel sheets now

- Rename the files to species.xlsx, facts.xlsx, stories.xlsx, collection.xlsx and put them in `backend/data/`.
- Species: fill the 7 missing ids and names; merge the 16 duplicate pairs and correct the 3 wrong rows (Appendix B); rewrite animal_class, department, iucn_status, diet, and development_type to the fixed lists; set every true/false column to T or F; map location and facility_name to the places list.
- Facts: give the 187 rows sharing id 1 new numbers; delete the 27 NULL rows and 12 duplicates; fix or delete the 58 rows whose species id matches nothing.
- Stories: fix or delete the 13 rows with a bad species id and the 1 empty row.
- Collection: rewrite all birth dates as mm/dd/yyyy with xx; set sex to male, female, or unknown; fill the 4 blank names, un-capitalize 68, and rewrite 4 nicknames as Name (Nickname); make species names match the species sheet; map habitats to the places list; rewrite previous institutions with only (birth) and (lived); add the public column.

The parser handles on every run

- Strips edge spaces, tabs, non-breaking spaces, and `_x000D_` markers.
- Accepts any capitalization in fixed-list columns and T/F columns.
- Splits a birth date into year, month, day, and the approximate flag; reads `xx` as unknown.
- Splits department lists and previous-institution lists on semicolons and reads the (birth) and (lived) tags.
- Separates a nickname from the name when written as Name (Nickname).
- Ignores StoriesID, Item Type, Path, and the priority value.
- Collapses identical story text across species into one shared passage, and records which Collection animals a story names.
- Builds every passage: species profiles, descriptions, facts, stories, animals, rosters. Leaves blank cells out of the text.
- Applies the public flag to passages, rosters, and stories.
- Checks everything else and refuses with a checklist instead of guessing.

> The dividing line: if a human could reasonably disagree about the right value, it is edited in Excel. If there is exactly one right answer and a machine can find it, the parser does it.

Birth dates

One written form for people, separate pieces for the computer. The sheet holds the written form; the ingest splits it into year, month, day, and an approximate flag, and anything that reads the date out loud follows the precision.

In the cell	Stored as	Read as
`10/03/2010`	2010, 10, 3	born October 3, 2010
`~06/16/1965`	1965, 6, 16, approximate	born around June 16, 1965
`10/xx/2020`	2020, 10, no day	born October 2020
`~xx/xx/1988`	1988, no month, no day, approximate	born about 1988
blank	nothing	birth date unknown

- Age and embargo math use the pieces. A missing month means "unknown", never a guess.
- Things like "+/- 1 month" or "arrived 02/02/2018, age unknown" go in Notes, not in the date cell.
- The sheet today has nine cells in this form and 193 that need rewriting. All 202 are listed in Appendix A.

What one ingest run does



- Run with one command from the project folder: `docker compose exec backend python -m app.features.knowledge.ingest`
- The checklist is a file with one line per problem: sheet, row, column, the value found, and what is allowed.
- If the checklist has even one line, nothing is loaded. The database keeps whatever it had before.
- When it loads, it prints the row count per table so you can compare with the sheet.
- Re-running after a small edit is quick: only passages whose text changed are embedded again.

How it is stored

Four tables mirror the sheets

- `species`, `facts`, `stories`, `animals`: one row in, one row out.
- Each keeps the original cell and the cleaned value side by side, so nothing is lost.
- Birth dates become year, month, day, and an approximate flag.

Two tables serve the search

- `chunks`: the passages, each with its source row, species, embedding, and visibility.
- `query_log`: every question asked, which passages were used, and the answer. The audit trail.

How passages are built

- Every passage starts with the species as "Common name (Scientific name)", so a question about elephants matches rows that never say the word.
- One species row makes one profile passage from its filled-in columns, in a fixed order: names and class, IUCN status, where it lives at the zoo, range and habitats, threats and conservation, development and lifespan, sizes, diets. Three columns stay out of the passage because they are links and citations: references, range_map, conservation_references.
- A species with a long description (151 of them) gets a second passage holding just the description.
- One fact makes one passage.
- One story makes one passage. When the same text sits under several species, the search sees one passage tagged with all of them, so it ranks once instead of 30 times. When a story names an animal that is in the Collection sheet, that link is recorded; 137 stories do, 217 do not, and that is fine.
- One animal makes one passage, with the birth date phrased by its precision and the notes included.
- One roster passage per species lists its animals with birth years, so "how many elephants do we have" has something to find.
- An animal with `public` = F is left out of public-facing passages, rosters, and the stories that name it. Staff still see everything.

Appendix A. The cleanup checklist

What the first ingest run will refuse, counted from the current files. Each line becomes an entry in the checklist file with the row number.

Sheet	What to fix	Cells or rows
species	Rows with no Title (1), no common name (1), no scientific name (5)	7
species	True duplicate species to merge (Appendix B), plus 3 rows with wrong or missing taxonomy	about 18
species	animal_class spelled 10 ways, mapped to 6	1,262 to review, few to change
species	department spelled 21 ways, mapped to 12; "None" becomes blank	as above

species	iucn_status outside the nine categories	12
species	diet outside the eight categories (menus move to diet_zoo)	25
species	development_type outside the five categories (details move to development_period)	26
species	True/false columns not T or F: venomous 41, sexually_dimorphic 56, ssp 48, handling 55, in_hzi_collection 6	206
species	location (77 spellings) and facility_name (45) mapped to the places list	as above
facts	FactsID 1 shared by many rows: renumber	187
facts	Summary is the word NULL: delete	27
facts	Same summary twice under one species: delete the copy	12
facts	species_id not in the species sheet (8 different ids, one of them 0)	58
stories	species_id blank or not in the species sheet	13
stories	Empty body	1
collection	Birth dates rewritten into mm/dd/yyyy with xx	193
collection	Sex not male/female/unknown (includes one cell reading "Peggy")	6
collection	House names: 4 blank, 68 in capitals, 4 nicknames to rewrite as Name (Nickname)	76
collection	Common or scientific name not matching the species sheet (Garbanzo the armadillo is filed under a bird)	16
collection	Current Habitat (42 spellings) mapped to the places list	202 to review
collection	Previous institutions: six tag spellings to two, and 22 "Born at Houston Zoo" cells rewritten as Houston Zoo (birth)	about 110
collection	Add the public column, all T	202

> Not on the list, on purpose: the 38 duplicated story rows across species (allowed), the 6 blank fact priorities (allowed), and the 6 house names shared across species such as the two Bobbis (allowed; the search will mention both).

Appendix B. Species rows to merge

The species sheet is two lists glued together: 513 hand-written signage rows with ids below 140,000, and 748 rows bulk-imported from the inventory with ids above. The pairs below are one species entered in both. Keep the first id, copy anything the second knew, delete the second. Nothing points at the deleted rows.

Species	Keep	Remove	Why they are the same
African lion	75	141437	two subspecies spellings of one lion
Fischer's turaco	227	141417	old and new genus name
Superb starling	73749	141026	Spreo and Lamprotornis are synonyms
Green jay	218	140729	identical scientific name
Vulturine guineafowl	95019	1235	identical scientific name, two exhibits
Red-billed blue magpie	3	141020	species and its nominate subspecies
Blue-crowned laughingthrush	141238	141000	synonyms
Red-fronted lemur	135216	140878	genus-only row; both have facts, merge them
Red-rumped agouti	143701	141472	species and subspecies
Mexican beaded lizard	303	141165	species and subspecies
New Caledonia giant gecko	307	141164	species and subspecies
South American bushmaster	141323	140993	species and subspecies
Leaf-nosed snake	141356	140985	genus-only row
Freshwater stingray	323	140942	genus-only row
Scarab beetle	144504	141549	genus-only row
Raccoon	141297	140879	second row has no scientific name

Three rows need a correction rather than a merge: the second "Red River hog" (141312) carries the scientific name of the Cape porcupine; the "Coral" row 140874 and the "Raccoon" row 140879 have no scientific name. The other 25 shared common names (nine "Coral", four "Brain coral", three "Barb", and so on) are different animals and stay as they are.

Appendix C. Fixed value lists

animal_class

- Mammal
- Bird
- Reptile
- Amphibian
- Fish
- Invertebrate

iucn_status

- Not Evaluated
- Data Deficient
- Least Concern
- Near Threatened
- Vulnerable
- Endangered
- Critically Endangered
- Extinct in the Wild
- Extinct

department

- Aquarium
- Birds
- Carnivores
- Children's Zoo
- Elephants
- Entomology
- Herpetology
- Hoofstock
- Natural Encounters
- Primates
- Sea Lions
- Wildlife Connections

diet and development_type

- diet: Carnivore, Herbivore, Omnivore, Insectivore, Frugivore, Piscivore, Granivore, Filter feeder
- development_type: Gestation, Incubation, Live birth, Ovoviviparous, Larval

Places (draft, for the zoo to confirm)

Built from the 42 habitat spellings in the Collection sheet. The species sheet's 77 location and 45 facility spellings get mapped to this same list during cleanup, and any place missing from it is added then. The gate refuses anything not on the list, which is how the list stays complete.

- African Forest
- African Forest - Giraffe
- Bears
- Birds of the World - Africa Aviary
- Birds of the World - North America Aviary
- Birds of the World - South America Aviary
- Bug House
- Children's Zoo
- Children's Zoo - Coastal Forest
- Children's Zoo - Desert Prairie
- Children's Zoo - Farm
- Children's Zoo - Forest
- Children's Zoo - Goat Yard
- Children's Zoo - Green Barn

- Duck Lake
- East Hoof Run
- Elephants (McNair Asian Elephant Habitat)
- Galapagos - Jelly
- Galapagos - One Ocean
- Galapagos - Penguins
- Galapagos - Sea Lion
- Galapagos - Tortoise
- Gorilla Exhibit
- Herpetology
- Hippo
- Kipp Aquarium
- Lion
- Natural Encounters - Ambassador
- Natural Encounters - Meerkat
- Natural Encounters - Rainforest
- Pantanal
- Pantanal - Canopy
- Pantanal - Cliffs Aviary
- Pantanal - Rainforest
- Pantanal - River
- Pantanal - Savanna Aviary
- Pygmy Hippo
- Red Panda
- Small Cat Row
- Texas Wetlands
- Tiger
- Tiger Building
- Tiger Building - Leopard Exhibit
- Tortoise Exhibit (Ankole)
- West Hoof Run
- White Rhino and Kudu Exhibit
- Wildlife Connections
- Wortham World of Primates (WWP)

Examples of what collapses: "East Hoofrun" and "East Hoof Run"; "CZ Farm", "CZ - Farm" and "Children's Zoo - Farm"; "NE-Meerkat" and "Natural Encounters - Meerkat"; "Galapagos - One Ocean Aquarium" and "Galapagos - One Ocean".

Still open

- What the fact priority number means. Stored, not used, until someone says.
- 41 species have a blank venomous cell. They become F in the cleanup; the zoo should glance at the list once, since a wrong F there matters.
- Whether Murphy (gibbon, born April 2026) and Tuve (howler monkey, born March 2026) are announced. Both get public = T unless the zoo says otherwise.
- The places list above needs a yes from someone who knows the grounds.